Appendix A - Beagle Marine Park Natural Values Report

[TEMPLATE] Marine Parks Network

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Natural Values

Understanding of natural values

Sampling summary

Table 1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. NPZ = National Park Zone, MUZ = Multiple Use Zone.

					Habitat	Fish	
Date	**Campaign name**	**Method**	**Areas**	**Samples**	**Count**	**Count**	**Length**
Sep-Oct 2018	201810 Beagle AMP stereoBRUV	BRUV	Coastal waters, MUZ, NPZ	127	127	127	117
Jan-Mar 2025	202501 Beagle AMP stereoBRUV	BRUV	Coastal waters, MUZ, NPZ	314	306	307	268

Benthic ecosystem extent

Observations

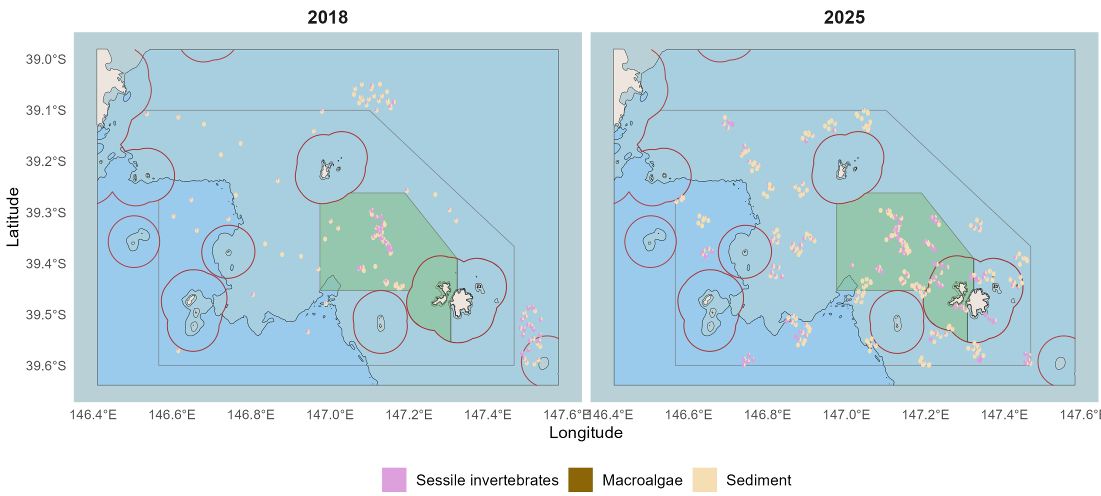

Figure 1: Habitat classes observed in spatially balanced stereo-BRUV imagery visualised as spatial pie charts. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 and 70 m is shown.

Dominant habitat types

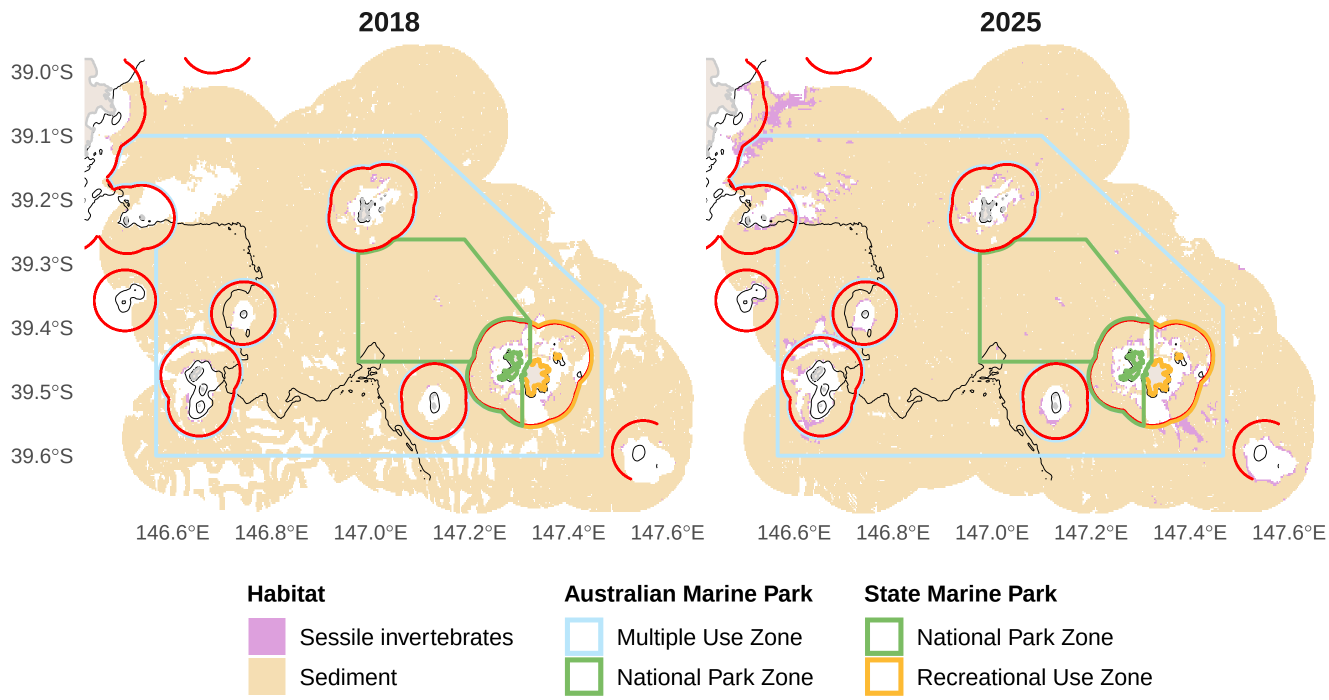

Figure 2: Predicted dominant habitat type from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 and 70 m is shown.

Habitat distribution and probability

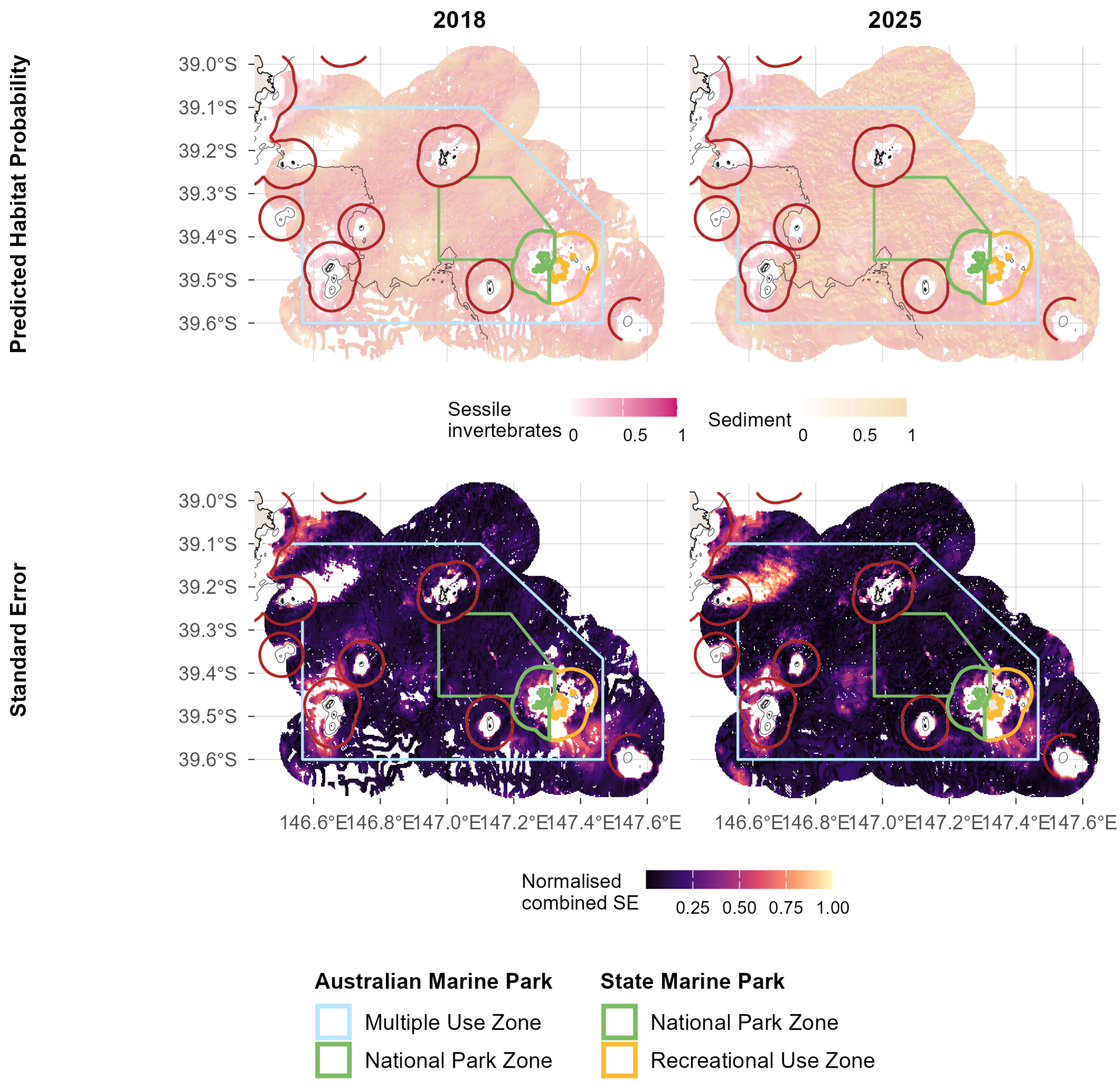

Figure 3: Predicted habitat distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 and 70 m is shown.

Functional reef distribution and probability

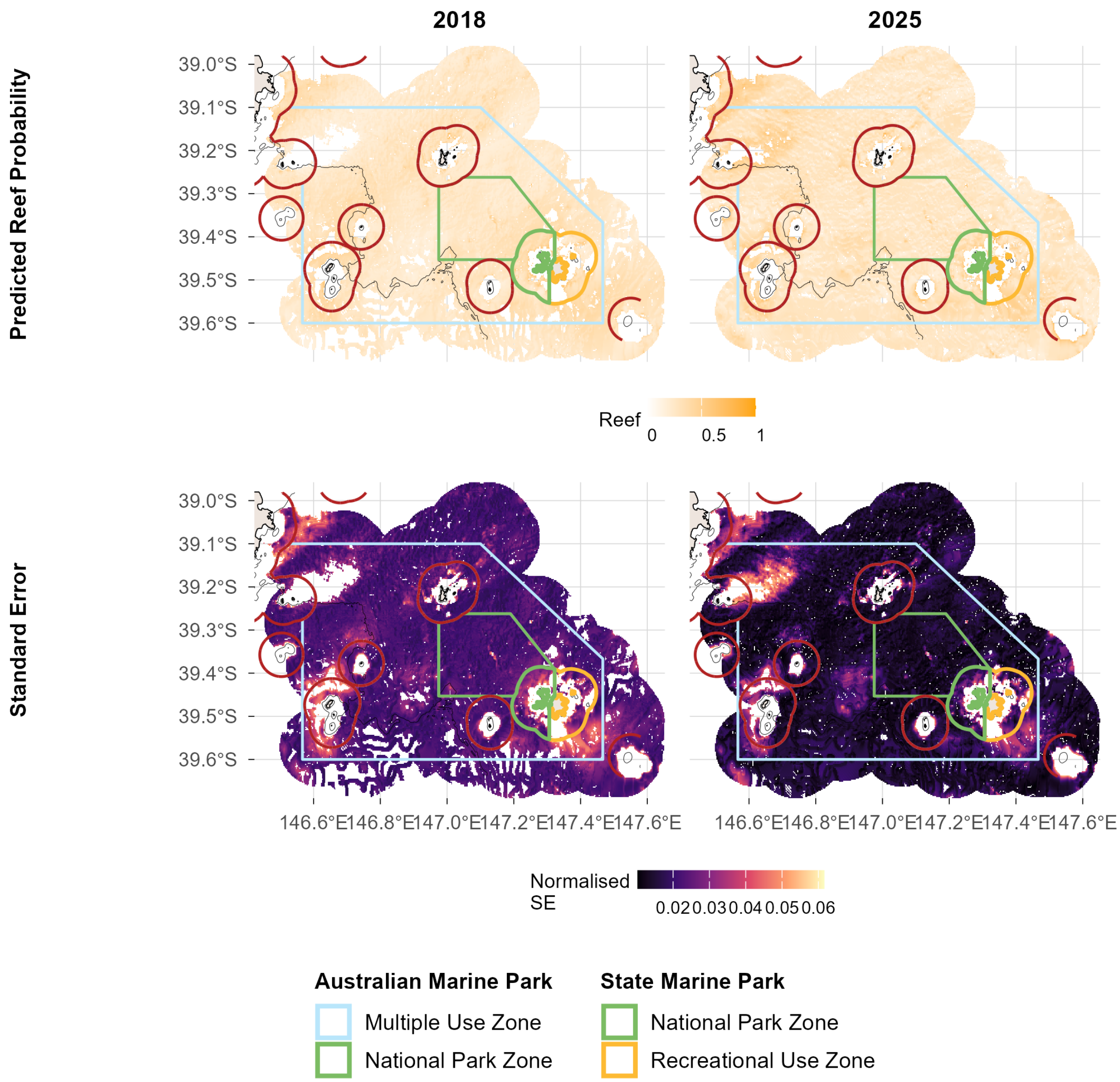

Figure 4: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 and 70 m is shown.

Benchmarks of benthic natural values

Sediment

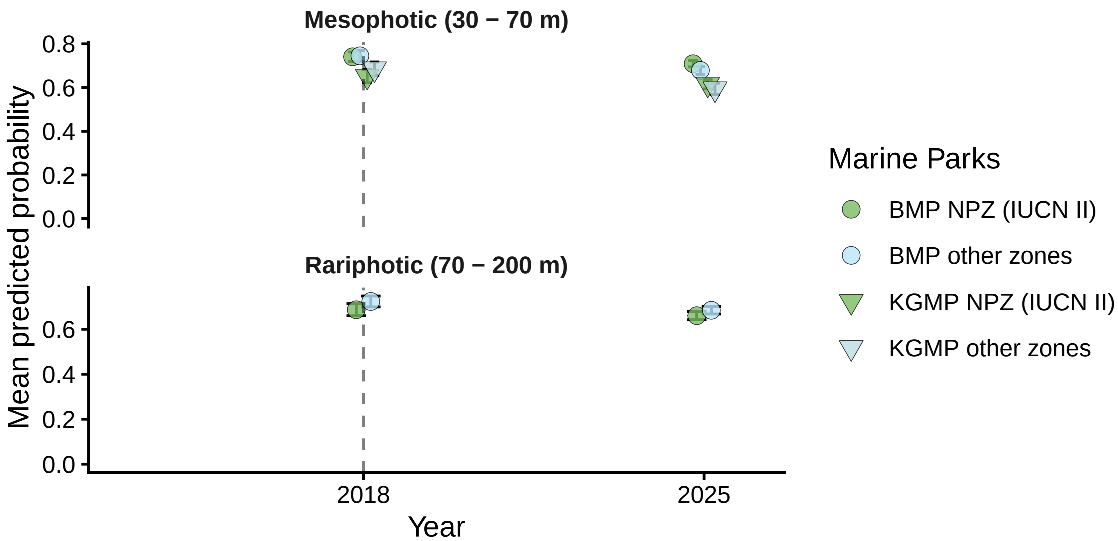

Figure 5.1: Sediment: temporal plots by depth contour from stereo-BRUV groundtruthing. BMP = Beagle Marine Park (Commonwealth), KGNP = Kent Group National Park (State).

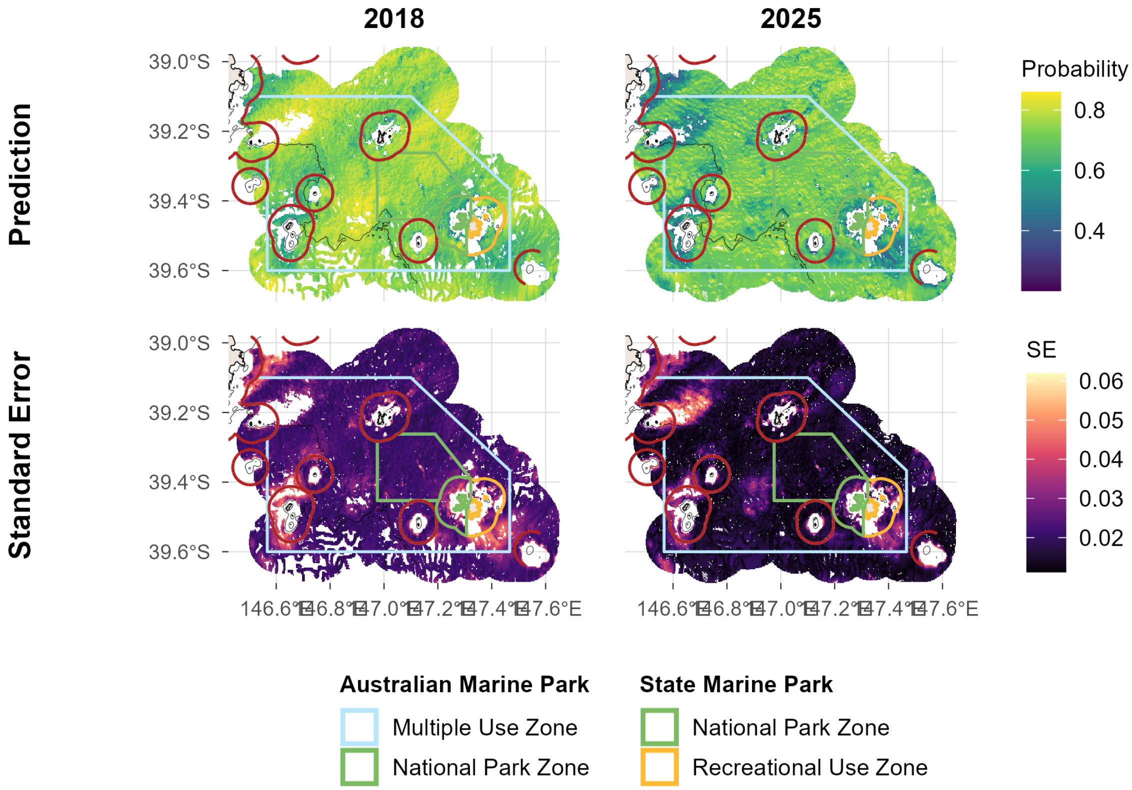

Figure 5.2: Sediment: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 and 70 m is shown.

Sessile invertebrates

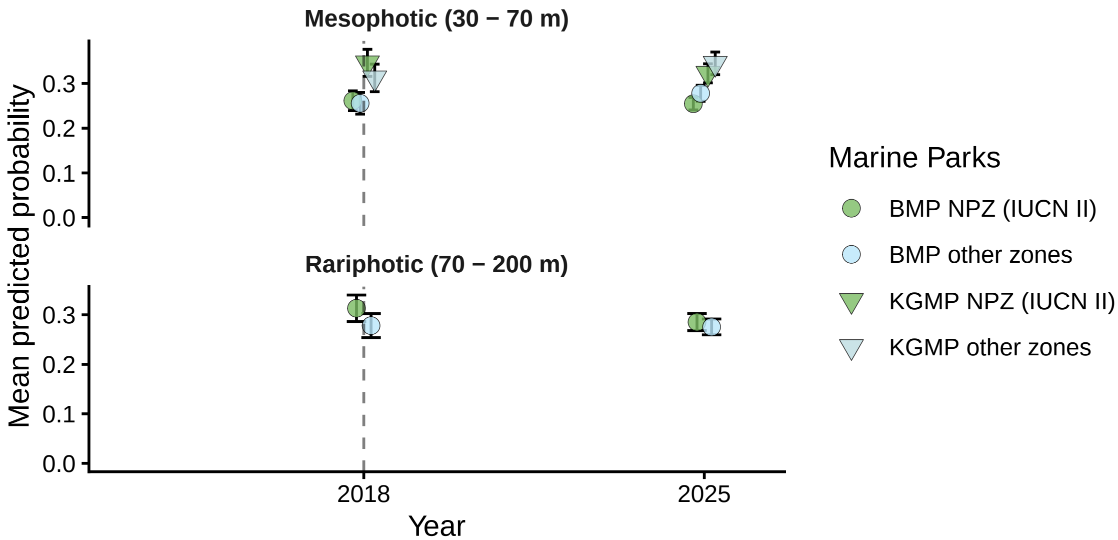

Figure 6.1: Sessile invertebrates: temporal plots by depth contour from stereo-BRUV groundtruthing. BMP = Beagle Marine Park (Commonwealth), KGNP = Kent Group National Park (State).

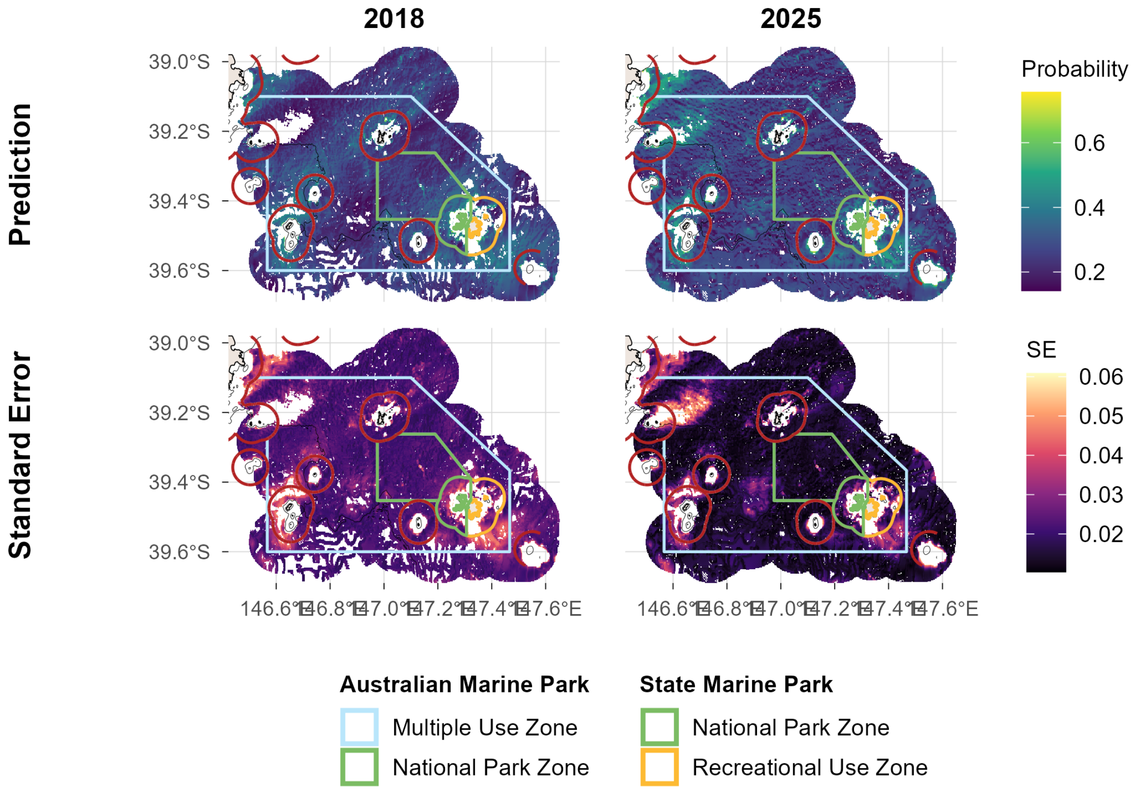

Figure 6.2: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 and 70 m is shown.

Benchmarks of demersal fish natural values

Survey effort

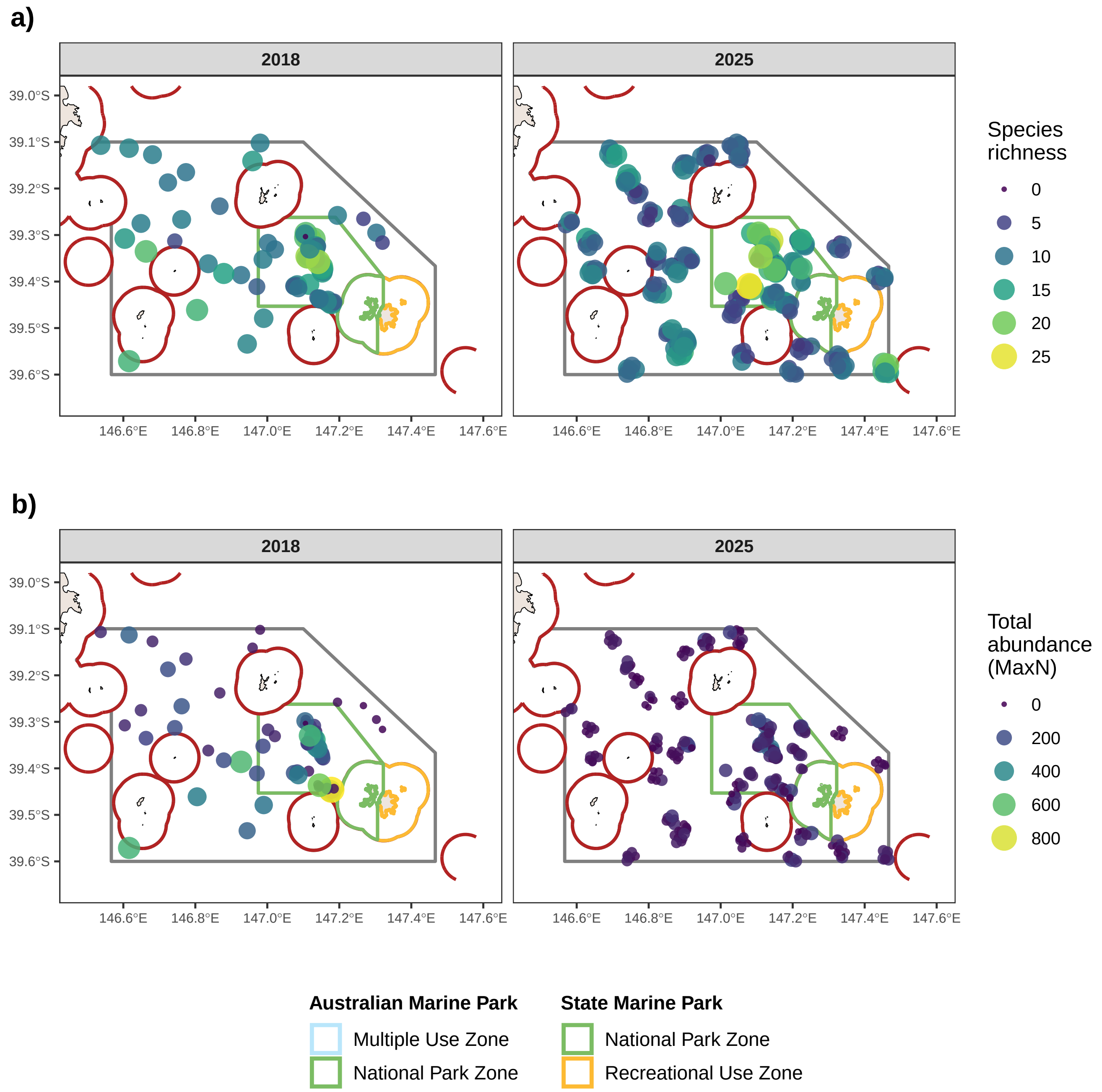

Figure 7: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

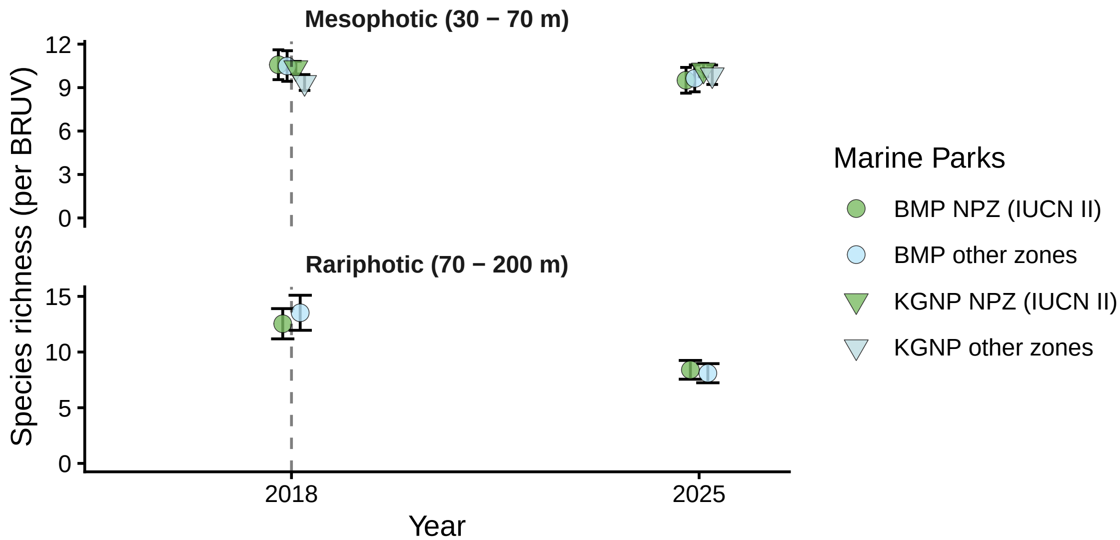

Figure 8.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. BMP = Beagle Marine Park (Commonwealth), KGNP = Kent Group National Park (State).

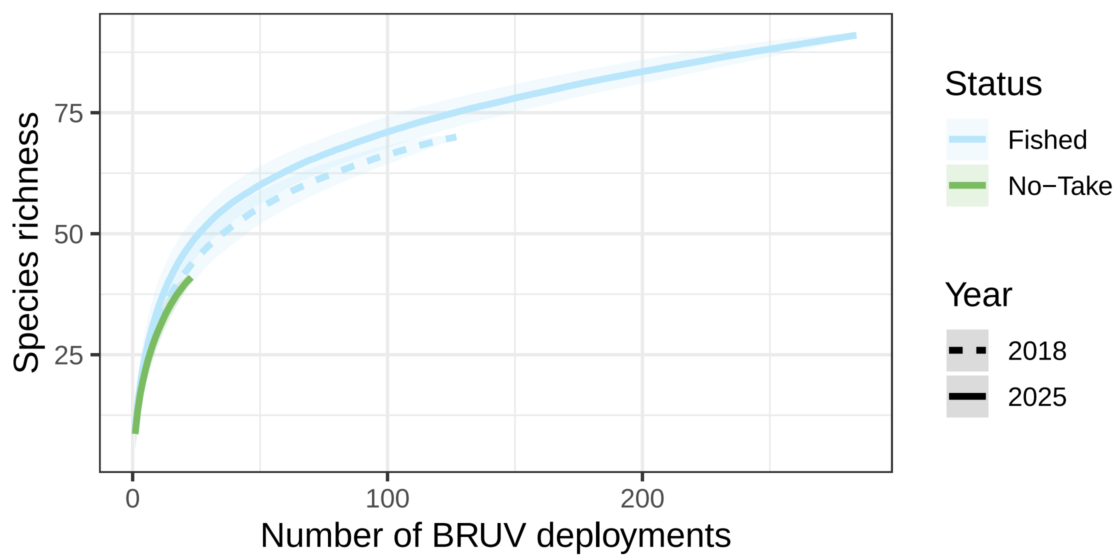

Figure 8.2: Demersal fish - Species richness: species accumulation curve by year from stereo-BRUV groundtruthing.

Species richness

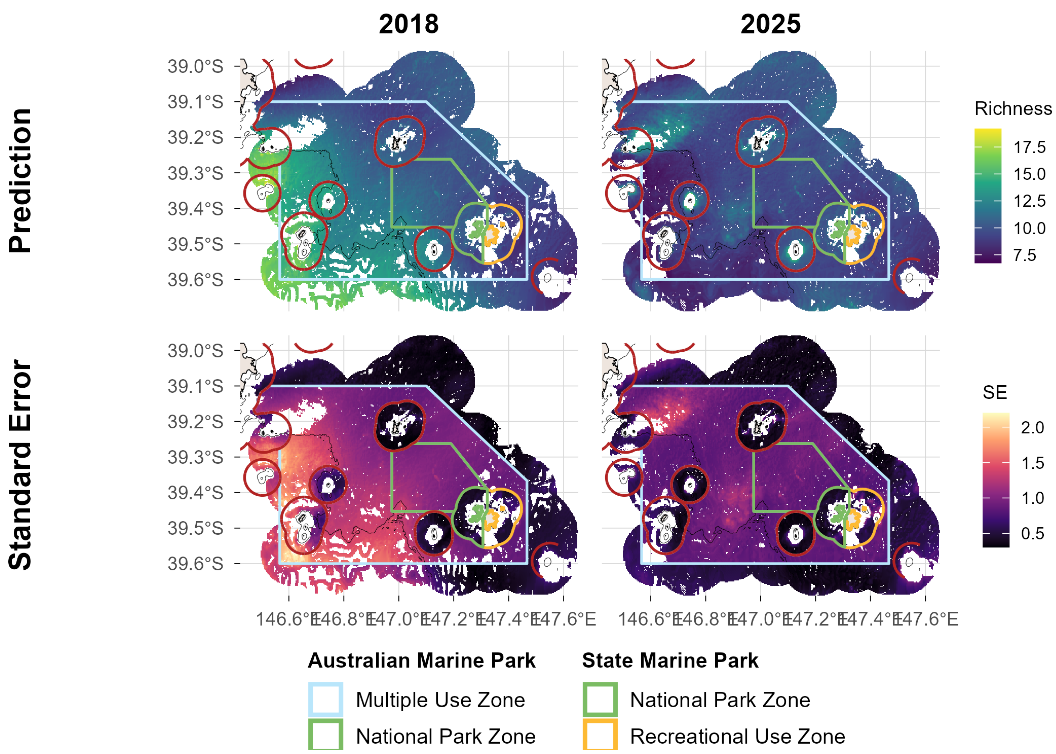

Figure 8.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 and 70 m is shown.

Total abundance

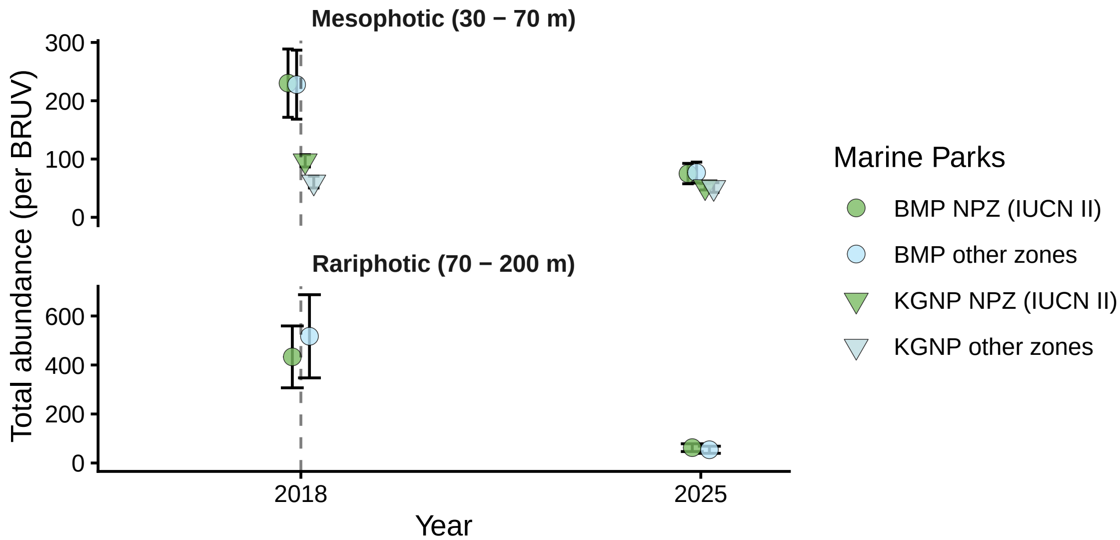

Figure 9.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. BMP = Beagle Marine Park (Commonwealth), KGNP = Kent Group National Park (State).

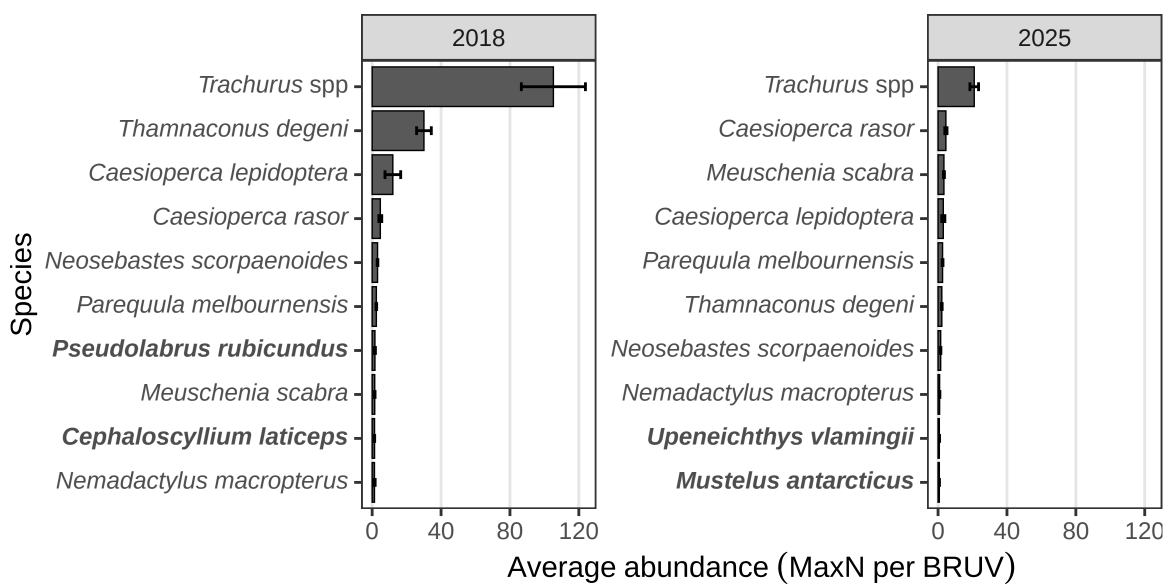

Figure 9.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing.

Total abundance

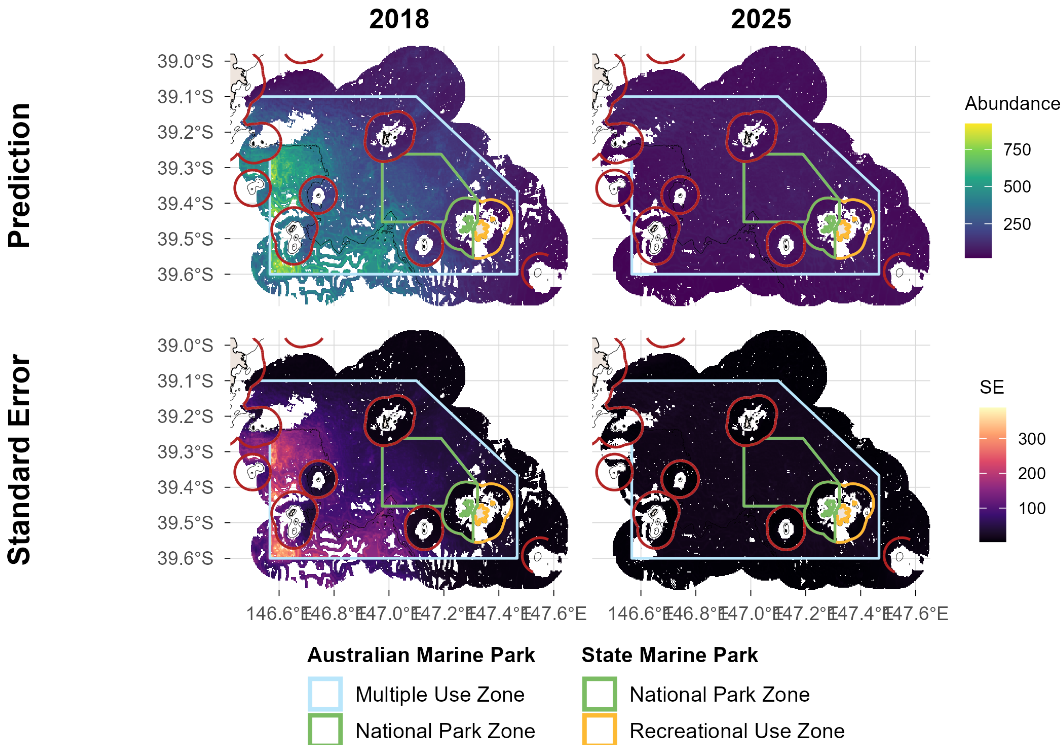

Figure 9.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 adn 70 m is shown.

Large Reef Fish Index*

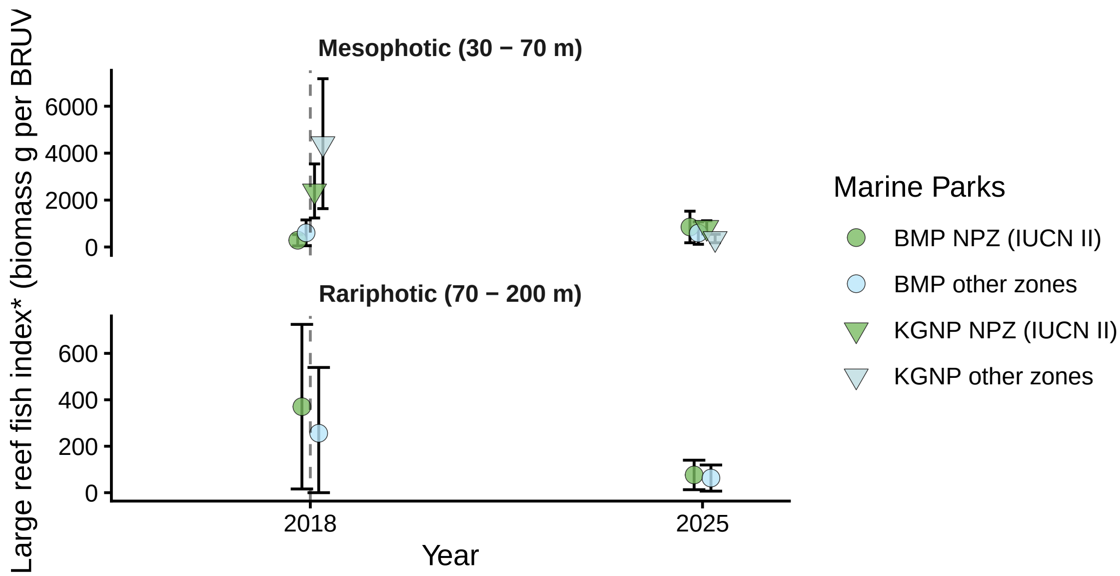

Figure 10.1: Demersal fish - Large Reef Fish Index*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. BMP = Beagle Marine Park (Commonwealth), KGNP = Kent Group National Park (State). *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

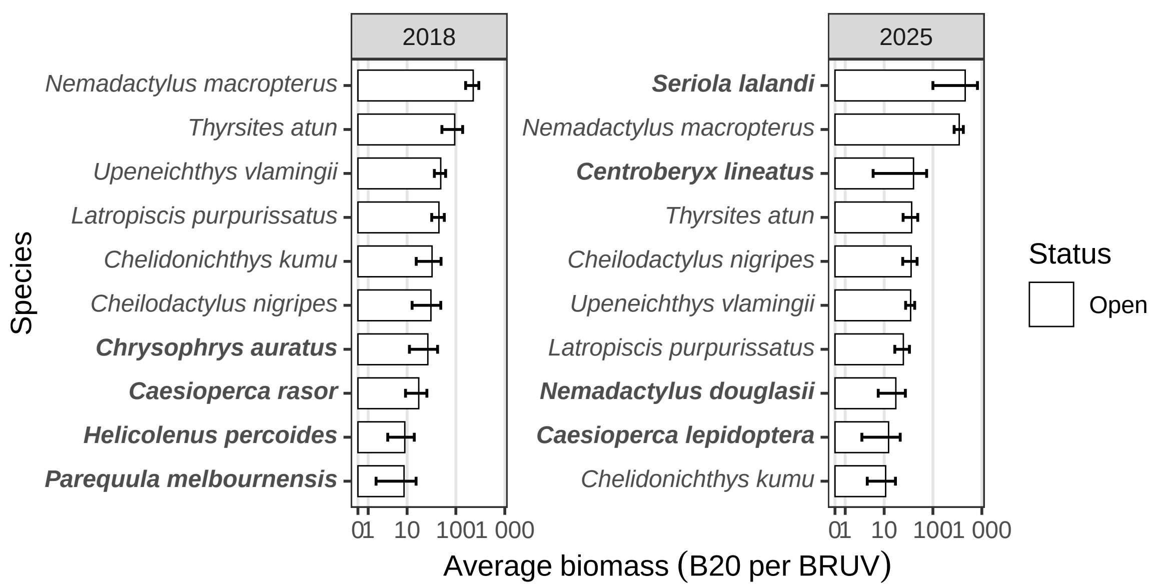

Figure 10.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species (pooled across deployments) from stereo-BRUV groundtruthing.

Large Reef Fish Index*

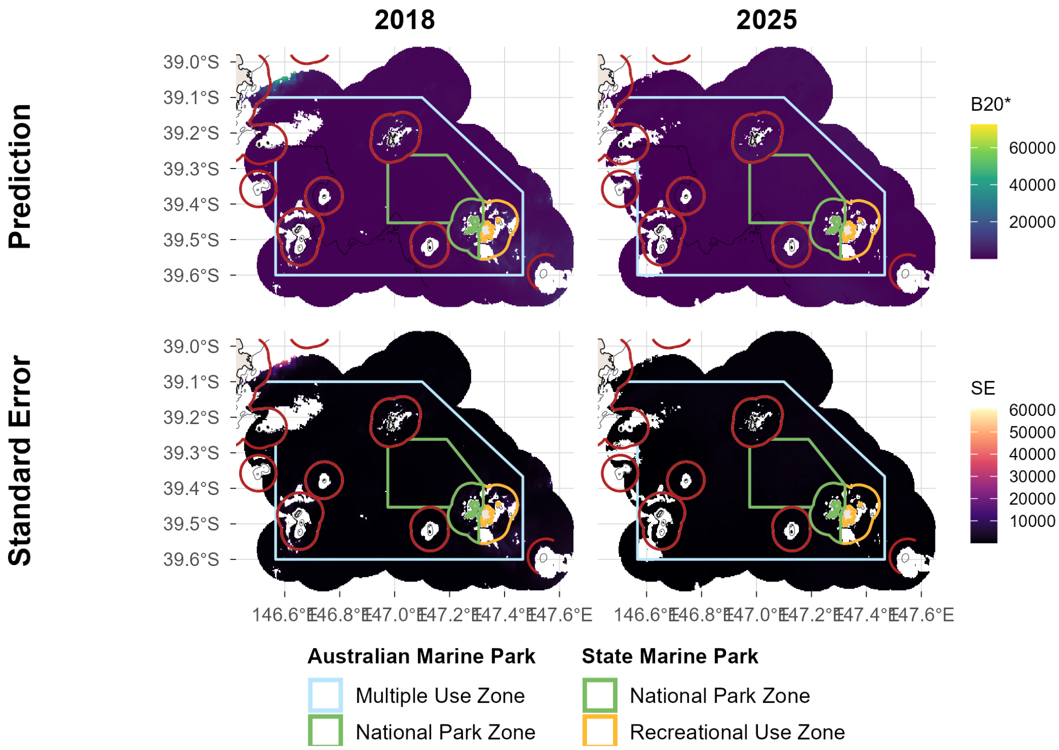

Figure 10.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 and 70 m is shown.

Reef fish thermal index

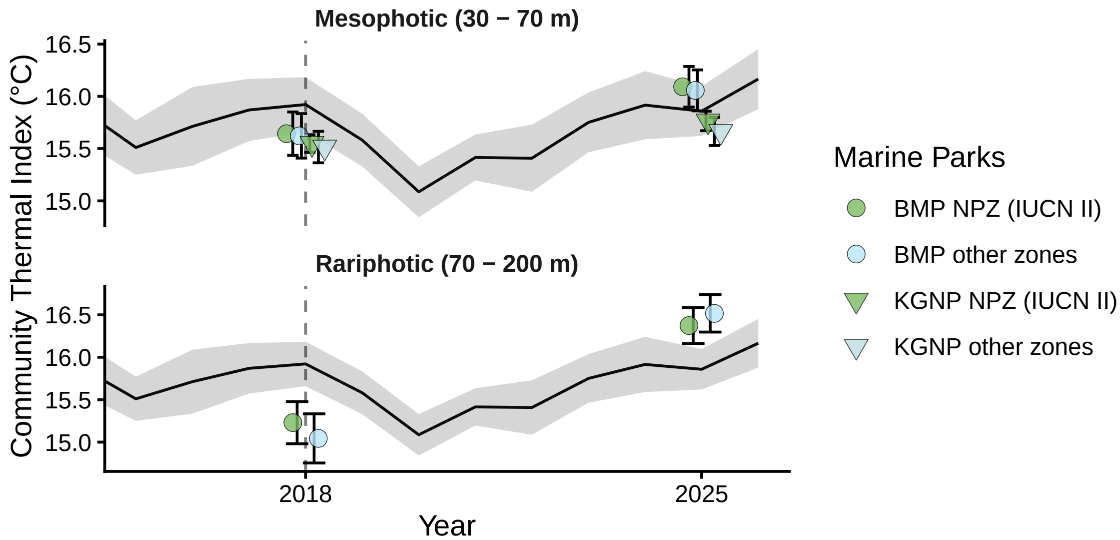

Figure 11.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. BMP = Beagle Marine Park (Commonwealth), KGNP = Kent Group National Park (State).

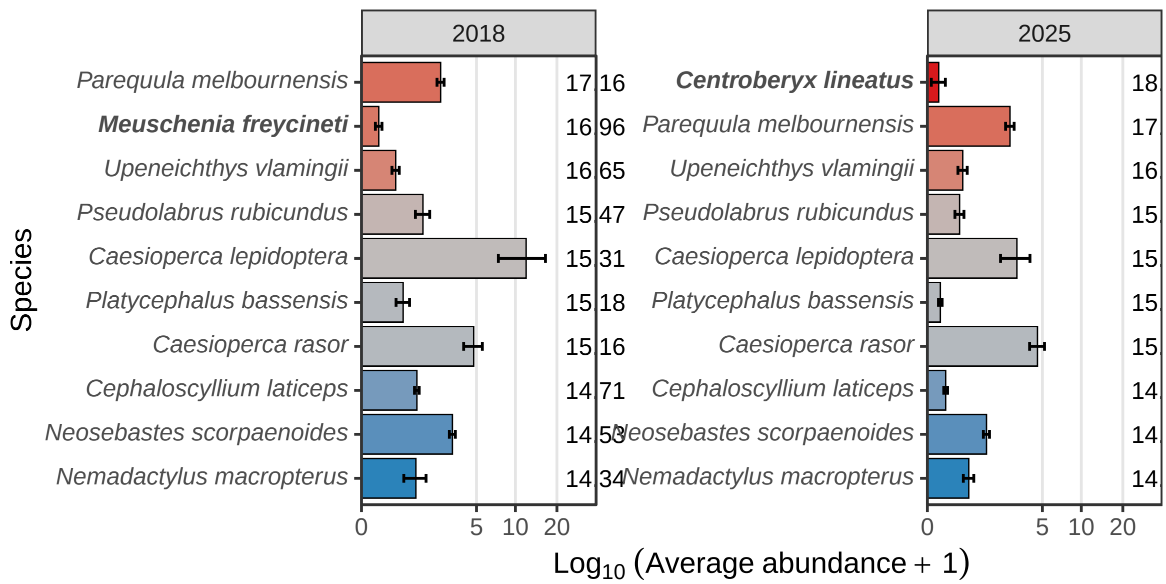

Figure 11.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing.

Reef fish thermal index

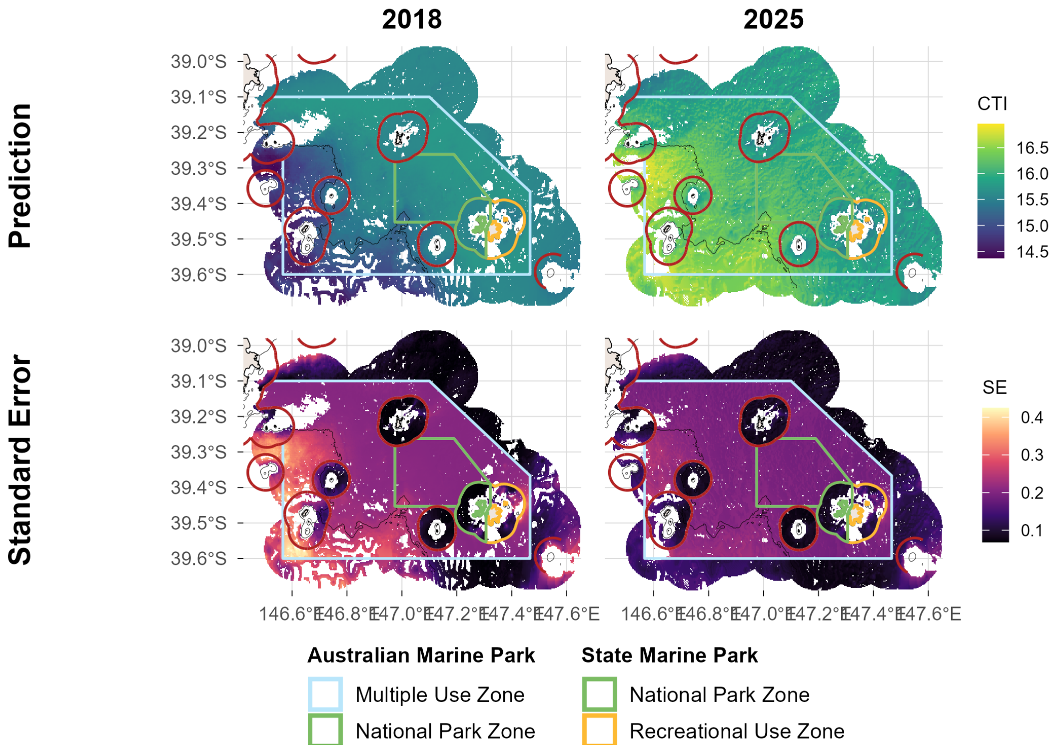

Figure 11.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth and State marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 and 70 m is shown.